R practical: worked solutions

Causal state-dependent local projections

DG ECFIN, 19 November 2026

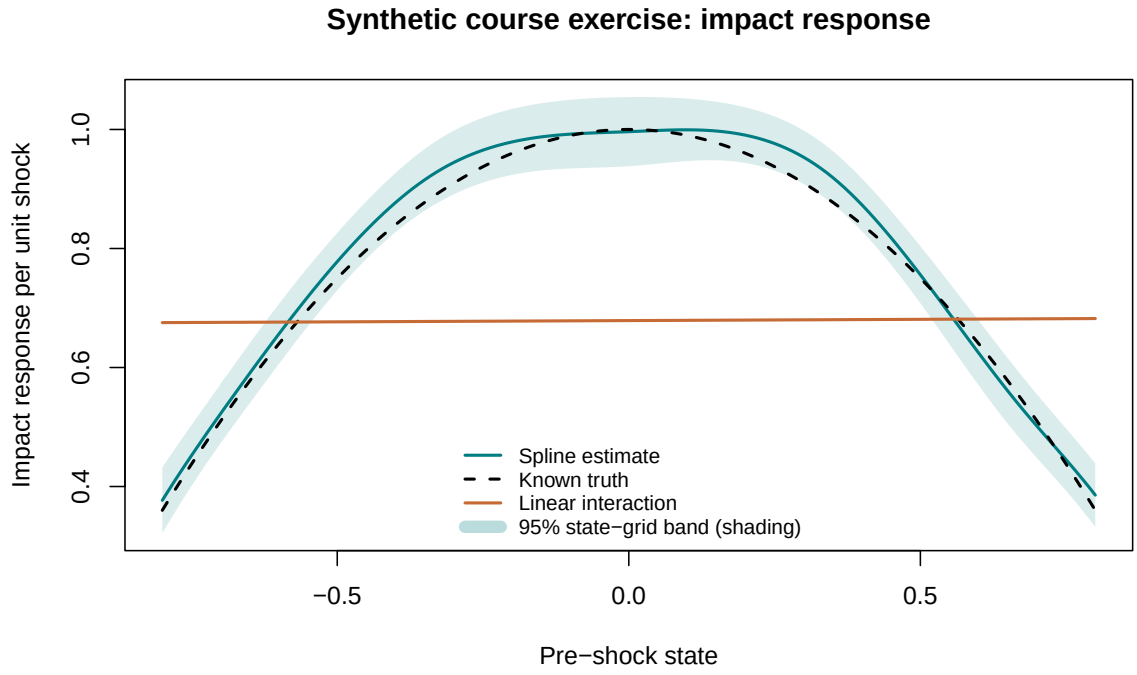
Run: `source("exercises/instructor_solutions.R")` from the course root. These answers use the supplied seed, baseline simulation, $J = 4{:}10$ and 1,999 Gaussian draws. Small numerical differences can arise across R versions.

Task 1: sample and timing

There are 24,000 rows, 100 units and 240 shock dates, with unique unit-date keys. The shock is common within date. The variable `z` already stores $Z_{i,t-1}$ and must not be lagged again. Cross-sectional replication does not create extra independent aggregate shocks. Common disturbances and overlapping-horizon errors motivate date-level score aggregation and HAC inference.

Task 2: fitted response and benchmark

The known impact response is $g_0(z) = 1 - z^2$. AIC over 4:10 selects $J = 9$ in this draw, with bandwidth 4. The linear response is approximately $0.6788 + 0.0042z$, which misses the hump. In population with a uniform symmetric state, the linear projection is the constant 2/3. A near-zero linear interaction therefore does not establish homogeneity.



The shading is a 95% state-grid Gaussian band at impact. This synthetic example illustrates the procedure; one realisation is not a coverage study.

Task 3: contrast and uncertainty

The exact impact contrast is $g_0(0) - g_0(.75) = .75^2 = .5625$.

Specification	Estimate	SE	Lower	Upper
AIC J=4:10	0.5514	0.0139	0.5241	0.5786
Fixed J=7	0.5724	0.0122	0.5484	0.5964

Both intervals in this draw include the truth. The variance of a difference includes the covariance between the fitted responses. Treating the two fitted states as independent gives the wrong formula.

At $z = 0$	Lower	Upper
Pointwise 95% interval	0.9523	1.0404
95% state-grid band	0.9382	1.0544

The Gaussian maximum gives critical value 2.586, compared with 1.960 pointwise. The band addresses all 161 grid points in $[-.8, .8]$ at $h = 0$. It is not joint across horizons. Coverage for the true function additionally requires bias control and regularity; AIC alone does not establish these conditions.

Task 4: sensitivity and interpretation

The fixed-$J = 7$ contrast is .5724, compared with .5514 for the selected basis. Its standard error is .0122 rather than .0139. Both the estimate and uncertainty can change when the basis changes. **Example paragraph.** In the synthetic panel, a unit shock has an estimated impact response .5514 larger at initial state zero than at state .75. The pointwise 95% interval for this prespecified contrast is [.5241,.5786]. The known contrast is .5625. The fitted response is hump-shaped; the nearly flat linear benchmark misses this heterogeneity. These findings concern the simulated design, rather than an empirical policy estimate. For a real application, justify shock exogeneity and initial-state timing. For inference, assess common and temporal score dependence, basis bias and the number of shock dates. For generated shocks or IV designs, additional inference is required.

Optional extensions

A. Horizon four. The exact contrast is $.6^4 \times .5625 = .0729$. The balanced panel retains 236 dates and 23,600 rows. Report pointwise intervals as such; the collection is not a joint band for the horizon path.

B. Bandwidth. Holding J fixed, changing the HAC bandwidth leaves the coefficient estimates unchanged and changes the covariance. This is a useful computational check.

C. Linear benchmark. The slope is close to zero because rising and falling parts of the response cancel in the projection. A flexible curve and prespecified contrasts are more informative about this shape.

D. Policy design. An adequate response names an outcome, intervention, initial state and horizon, plus a concrete exogeneity argument and threat. Merely listing controls is insufficient.

E. Endogenous state. The state transition depends on past shocks by construction, but the stored state precedes the current shock. Constant propagation and independent future innovations preserve the affine conditional mean. Conditioning on the post-shock state would answer a different question and may block transmission or induce selection.

F. Nonlinear shock exposure. At $z = 0$, $\delta = 1$ and cubic coefficient .25, the LP slope times the shift is 1.75 and the finite-shift target is 2. Their discrepancy is .25. This is the omitted $c\delta^3$ term. A spline in the state cannot remove a misspecification in the shock dimension.

Files produced by the solution script

The directory `output_R_paper/` contains:

- `core_answers.csv` and `core_summary.csv`
- `core_curve.csv` and `linear_benchmark.csv`
- `core_comparison.pdf`

The full demonstration also reports horizon curves, state contrasts and bandwidth sensitivity. All results use simulated data.